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# MILITARY SPECIFICATION

## REGULATOR ASSEMBLY, CARBON DIOXIDE PRESSURE

This Specification is approved for use by all Departments And Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of carbon dioxide pressure regulator assembly referenced herein as the "regulator".

## 2. APPLICABLE DOCUMENTS

* 2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging, Methods of |
| MIL-F-18866 | Fittings, Hydraulic Tube, Flared, 37-Degree And Flareless; Steel |
| MIL-S-22473 | Sealing, Locking, and Retaining Compounds; Single-Component |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |

FSC 1355

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| | |
|---|---|
| MS-19060 | Ball, Bearing, Ferrous, Corrosion Resisting Steel |
| MS-20995 | Wire, Safety or Lock |

DRAWINGS

Naval Ordnance Systems Command

| | |
|---|---|
| DL-1330653 | Regulator Assembly |
| LM-2526799 | Test Assembly, Regulator |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC.

| | |
|---|---|
| NAS 617 | Packing, Preformed, Straight Thread Tube Boss, Synthetic Lubricant Resistant |

(Application for copies should be addressed to Aerospace Industries Association of America, Inc., 1725 DeSales St. N.W., Washington, D.C. 20036)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

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3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the regulator are classified herein as follows:

| REQUIREMENT | PARAGRAPH |
|---|---|
| General Requirements | 3.2 |
| Performance Requirements and Product Characteristics | 3.3 |
| Environmental Requirements | 3.4 |
| Operational Service Life | 3.5 |

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the regulator is required and shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract (see 4.3.1).

* 3.2.2 Materials. Unless otherwise specified in the contract or order (see 6.2), all materials and treatments of materials and parts used in the manufacture of the regulator shall be in accordance with applicable documents listed on DL1330653.

3.2.3 Weight. The weight of the regulator shall not exceed one pound.

3.2.4 Definitions. See 6.3 for definitions of terms.

3.3 Performance requirements and product characteristics. Unless specified otherwise (see 6.2), the regulators shall satisfy the following performance requirements when tested as specified in Section 4:

* 3.3.1. Pressure regulation. The regulator, connected to the Regulator Test Assembly, LM 2526799, shall maintain an outlet pressure between 80 and 130 pounds per square inch, gage (psig) with an inlet pressure from 250 to 2700 psig, with valve "A" in small orifice position (See Figure 1).

* 3.3.2. Flow. The regulator, connected to the Regulator Test Assembly, LM 2526799, with an applied inlet pressure of 250 ±10 psig shall maintain an outlet pressure greater than 40 psig with valve "A" in large orifice position (See Figure 1).



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3.3.3 Relief valve. The relief valve shall have a cracking pressure of 200 ±25 psig and a reset pressure of 140 psig minimum. At 140 psig, the valve shall have a maximum leakage rate of 0.5 cubic centimetres per minute at Standard Test Conditions ($cm^3$/min-Std).

3.3.4 Case leakage. The regulator assembly, in the operating condition, shall show no sign of leakage when tested in accordance with 4.5.3(c).

* 3.3.5 Check valve. The check valve shall have a cracking pressure of 4 ±2 psig. With 300 ±25 psig at the outlet there shall be no external or reverse flow leakage as detectable by a bubble test. Similarly, there shall be no reverse flow or external leakage when a pressure of 12 ±2 inches of water is applied to the outlet of the check valve.

3.4 Environmental requirements. Unless otherwise specified (see 6.2), the regulator shall be capable of conforming to the performance requirements and product characteristics of this specification after exposure to the following environmental conditions (Ready, operating, and non-operating are defined in 6.3)

3.4.1 Vibration, ready condition. The regulator, while in the ready condition and mounted as shown in figure 2, shall be capable of meeting the performance requirements of 3.3 after being subjected to the vibrations specified in Table I. The regulator shall be vibrated for 180 minutes each in the longitudinal axis and in the two mutually perpendicular transverse axes (see figure 2), at various temperatures within the range of -43 to 60°C (-45 to 140°F) (vibration is defined in 6.3.4) (see 4.4.4.2).

3.4.2 Acceleration, ready condition. The regulator, in the ready condition and mounted as shown in Figure 2, shall be capable of meeting the performance requirements of this specification after being subjected to the accelerations specified in Table II. (Acceleration is defined in 6.3.5).

TABLE I

Vibration Schedule

| Double Amplitude (in. ±10%) Minimum | Acceleration (g's ±10%) Minimum | Frequency (Hz) |
|---|---|---|
| 0.2 | --- | 5 - 22 |
| --- | 3.0 | 22 - 40 |
| 0.036 | --- | 40 - 74 |
| --- | 10 | 74 - 500 |



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TABLE II

Acceleration Schedule

| Magnitude (g's ±10%) | Acceleration Direction (See Figure 2) | Duration |
|---|---|---|
| 25 | Longitudinal axis, Fwd | 1 minute |
| 35 | Longitudinal axis, Aft | 1 minute |

3.4.3 Shock, ready condition. The regulator, in the ready condition and mounted as shown in Figure 2, shall be capable of meeting the performance requirements of this specification after being subjected to the shock loads specified in Table III (Shock is defined in 6.3.6).

3.4.4 Temperature, nonoperating. The regulator, in the nonoperating condition, shall be capable of withstanding temperatures between -43 and 60°C (-45 and 140°F) (Temperature is defined in 6.3.7) (see 4.4.4.5).

3.4.5 Humidity, nonoperating. The regulator, in the nonoperating condition, shall be capable of withstanding an environment of at least 95 percent relative humidity at 43°C (110°F) for a minimum of 8 hours (humidity is defined in 6.3.8) (see 4.4.4.5).

* 3.4.6 Temperature, ready, operating. The regulator, immediately after conditioning for 1 hour at any temperature between -18 and 60°C (0 and 140°F), shall be capable of meeting the performance requirements while at any temperature between 0 and 60°C (32 and 140°F) (see 4.4.4.6 and 4.4.4.7).

3.5 Operational service life. The regulator shall have a service life of 200 minutes when tested in accordance with 4.4.4.10.

3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.



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TABLE III

Shock Test Schedule

| Peak Magnitude (g's ±10%) | Duration (ms) | Impact Axis | Direction of Deceleration Force | Number of Impacts |
|---|---|---|---|---|
| 300 | 0.5 to 1.5 | Longitudinal | Aft | 3 |
| 85 | 0.5 to 1.5 | Transverse A | Optional | 3 |
| 85 | 0.5 to 1.5 | Transverse B | Optional | 3 |
| 45 | 18.0 to 20.0 | Longitudinal | Fwd | 3 |
| 60 | 8.0 to 10.0 | Longitudinal | Aft | 3 |
| 60 | 8.0 to 10.0 | Longitudinal | Fwd | 3 |
| 60 | 8.0 to 10.0 | Transverse A | Optional | 3 |
| 60 | 8.0 to 10.0 | Transverse B | Optional | 3 |

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the regulator shall be classified as follows:

(a) Preproduction inspection (see 4.4)

(b) Quality conformance inspection (see 4.5)

(c) Periodic production inspection (see 4.6)

(d) Examination of first echelon repair parts (see 4.7)



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4.3 Sampling

* 4.3.1 Preproduction inspection sample. A preproduction sample of 6 regulators manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests, listed in Table IV and detailed in 4.4, at an activity designated by the procuring activity (see 6.2). Failure of a regulator to successfully pass any of the tests in 4.4 shall be cause for rejection of the preproduction sample. Further production of the regulator by the supplier, prior to approval of the preproduction sample shall be at the supplier's risk.

4.3.2 Periodic production sample. Periodic production samples shall be randomly selected from units that have satisfactorily passed the quality conformance inspection of 4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. (see 6.2). Periodic production sampling shall be as follows:

Step 1. Upon initiation of production, randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Examination of first echelon repair parts. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of first echelon repair parts for the regulator as specified in 4.7.

4.4 Preproduction inspection

4.4.1 Tests and examinations. The preproduction sample of 6 regulators shall be inspected in accordance with Table IV, with 2 regulators tested in accordance with each of the 3 test sequences; A, B and C.



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TABLE IV

Preproduction Test Sequence

| Test | Paragraph Reference | Test Sequence | | |
|---|---|---|---|---|
| | | A | B | C |
| Operational Test | 4.4.4.1 | 1 | 1 | 1 |
| Vibration, Ready Condition | 4.4.4.2 | 2 | | |
| Acceleration, Ready Condition | 4.4.4.3 | | 2 | |
| Shock, Ready Condition | 4.4.4.4 | | 3 | |
| Operational Test | 4.4.4.1 | 3 | 4 | |
| Temperature and Humidity | 4.4.4.5 | 4 | 5 | |
| Low Temperature Operating | 4.4.4.6 | 5 | | |
| High Temperature Operating | 4.4.4.7 | | 6 | |
| Vibration, Shipping | 4.4.4.8 | | | 2 |
| Shock, Shipping | 4.4.4.9 | | | 3 |
| Operational Test | 4.4.4.1 | | | 4 |
| Operational Service Life | 4.4.4.10 | 6 | 7 | 5 |
| Operational Test | 4.4.4.1 | 7 | 8 | 6 |

4.4.2 Test conditions. Unless otherwise specified, (see 6.2) the regulator shall be subjected to the preproduction inspection under the following conditions:

(a) Temperature: Room ambient 0 to 37°C (32 to 98°F)

(b) Altitude: Normal ground atmospheric pressure

(c) Vibration: None

(d) Humidity: Room ambient



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4.4.3 Test equipment and setup. In addition to the items listed in 4.5.2, the following test equipment is required:

4.4.3.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

(a) Frequency range, logarithmic sweep: 5 to 500 Hertz (Hz)

(b) Test specimen: 1 pound

(c) Acceleration and displacement (ready condition):

| | |
|---|---|
| 5 to 22 Hz | 0.2 inch |
| 22 to 40 Hz | 3.0 gravitational units (g's) |
| 40 to 74 Hz | 0.036 inch |
| 74 to 500 Hz | 10 g's |

(d) Acceleration (packed and packaged):

| | |
|---|---|
| 5 to 60 Hz | 1 g |

(e) Automatic cycling time range: 0 to 15 minutes (optional)

(f) Temperature chamber range: -43 to 60°C (-45 to 140°F) accuracy to ±3°C (±5°F)

(g) Interior size: 2.5 x 3.5 x 5.5 inches minimum

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with a 1-pound test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting a 1-pound test specimen and shall provide ranges of:

(a) 300 g's for 0.5 milliseconds (ms) to 1.5 ms

(b) 60 g's for 8.0 ms to 10.0 ms

(c) 45 g's for 18.0 ms to 20.0 ms

4.4.3.4 Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics:

(a) Temperature: -43 to 60°C (-45 to 140°F) with an accuracy to ±3°C (±5°F)

(b) Relative humidity: 95 percent minimum at 43°C (110°F), with an accuracy to ±3°C (±5°F)

(c) Interior size: 2.5 x 3.5 x 5.5 inches minimum.

This equipment may be used for both nonoperating and ready-operating temperature (and humidity) tests.



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4.4.4 Test procedures

4.4.4.1 Operational test. The regulator shall be subjected to the operational tests detailed in 4.5.3.

* 4.4.4.2 Vibration, ready condition. The regulator, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 2. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 Hz to 500 Hz. The regulator shall then be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table V. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -43 ±3°C (-45 ±5°F), room temperature (see 4.4.2), and 60 ±3°C (140 ±5°F), respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes specified in Table I.

TABLE V

Vibration Test Schedule

| Number of Resonances | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Total Vibration Time at Resonance * | - | 30 mins | 1 hr | 1-1/2 hrs | 2 hrs |
| Cycling Time | 3 hrs | 2-1/2 hrs | 2 hrs | 1-1/2 hrs | 1 hr |

*30 minutes at each resonance

* 4.4.4.3 Acceleration, ready condition. The regulator, in the ready condition and secured to an acceleration equipment (see Figure 2), shall be subjected to the accelerations specified in Table II. Acceleration shall be measured at the approximate geometric center of the regulator.

* 4.4.4.4 Shock, ready condition. The regulator, in the ready condition, shall be attached to a shock test machine (see Figure 2). Shock magnitude and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the regulator shall be performed in accordance with Table III. Test sequence is optional.



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4.4.4.5 Temperature and humidity, nonoperating condition. Four preproduction regulators shall be subjected, in the nonoperating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stablized at a temperature of -43 + 0, -3°C(-45 + 0, -5°F) and remain for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -43 + 0, -3°C (-45 + 0, -5°F) within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours). The specimen shall then be subjected to a temperature of 60 + 3, -0°C (140 + 5, -0°F). Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of 43 + 3, -0°C (110 + 5, -0°F) at a minimum relative humidity of 95 percent for a minimum of 8 hours. Prior to the tests of 4.4.4.6 and 4.4.4.7 the regulator may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.4.6 Low temperature operating. Before testing, the regulator, installed in its test chamber in the ready condition, shall be conditioned at a temperature of -18 ± 3°C (0 ± 5°F) for a period of 1 hour. Immediately thereafter, the regulator shall be operated in accordance with and meet the requirements of 4.5.3(a) at a temperature of 0 ± 3°C (32 ± 5°F). The requirements of 4.5.3(b), (c), and (d) shall be met at ambient temperature.

4.4.4.7 High temperature operating. Before testing, the regulator, installed in its test chamber in the ready condition, shall be conditioned at a temperature of 60 ± 3°C (140 ± 5°F) for a period of 1 hour. Immediately thereafter, the regulator shall be operated in accordance with and meet the requirements of 4.5.3(a) at a temperature of 60 ± 3°C (140 ± 5°F). The requirements of 4.5.3(b), (c), and (d) shall be met at ambient temperature.

4.4.4.8 Vibration, packed and packaged. The regulator, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 Hz. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes.



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4.4.4.9 Shock, packed and packaged. The regulator, packed and packaged in accordance with Section 5, shall be subjected to free fall drop tests. The regulator in its container shall be dropped from a height of 36 inches, one drop on each of 4 diagonally opposite corners and flat drops, one each, on the bottom and on one side. The drops shall be made on an impact surface of essentially nonyielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: room temperature, within 10 minutes after the container has been stabilized at -43 ± 3°C (-45 ± 5°F), and within 10 minutes after the container has been stabilized at 60 ± 3°C (140 ± 5°F).

* 4.4.4.10 Operational service life. Connect regulator to test equipment in accordance with Figure 1. Increase the supply pressure to 1500 ±50 psig and record the outlet pressure. This pressure shall be between 80 to 130 psig with valve "A" in small orifice position. Cycle the regulator at 1500 ±50 psig inlet pressure for 200 ±2 minutes, with valve "A" closed for 4.5 minutes and in small orifice position for 0.5 minute. After completion of cycling, outlet pressure shall be within ±15 psig of original recorded pressure, with inlet pressure maintained at 1500 ±50 psig with valve "A" in small orifice position.

4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier as specified in the contract or order (see 6.2) and maintained as part of the inspection records. In addition to the data requirements of 4.5.4, the data sheets shall contain, but are not limited to, the following information:

(a) Date_____

(b) Regulator, serial number_____

(c) Preproduction tests:

Low temperature operating
Temperature_____°C (°F)

High temperature operating
Temperature_____°C (°F)

(d) Operational life cycling:

| | | |
|---|---|---|
| Inlet pressure | _____ | psig |
| Outlet pressure | _____ | psig prior to cycling |
| On-time | _____ | Minutes |
| Off-time | _____ | Minutes |
| Total time | _____ | Minutes |
| Outlet pressure | _____ | psig after cycling |



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* 4.5 Quality conformance inspection. Each regulator produced under the contract or purchase order shall be subjected to the tests and examinations of 4.5.3. Failure of a regulator to successfully pass the tests in 4.5.3(a) and 4.5.3(c) shall be cause for rejection of that regulator. Failure of the relief valve or check valve under 4.5.3(b) and 4.5.3(d) will only reject these parts and not the entire assembly; except when acceptance test is part of preproduction or periodic testing.

4.5.1 Test conditions. Unless otherwise specified (see 6.2), the regulator shall be subjected to the quality conformance tests at the conditions specified in 4.4.2.

4.5.2 Test equipment and setup. The test equipment setup shall be in accordance with Figure 1.

(a) Inert gas supply: 0 to 2800 psig minimum

(b) Pressure gage: 0 to 3000 psig, accuracy ± 2 percent full scale

(c) LM 2526799: Regulator Test Assembly

* 4.5.3 Operational test procedures. (sequence optional)

(a) Pressure regulation and flow. Connect the regulator to the Regulator Test Assembly, LM 2526799, in accordance with Figure 1. With valve "A" in small orifice position, adjust the inlet pressure to 250 ± 10 psig. The outlet pressure shall be between 80 psig and 130 psig. Increase the inlet pressure to 1500 ± 50 psig. The outlet pressure shall not exceed 130 psig. Increase the pressure to 2700 ± 50 psig. The outlet pressure shall not exceed 130 psig. Decrease inlet pressure to 250 ± 10 psig with valve "A" in large orifice position. Outlet pressure shall be greater than 40 psig.

(b) Relief valve. Connect the relief valve to a regulated pressure source. Increase the inlet pressure. Cracking pressure shall be 200 ± 25 psig. Reduce the inlet pressure. Reset pressure to 140 ± 5 psig. Measure the leakage rate. The leakage rate shall not exceed 0.5 $cm^3$/min-Std. Replace the relief valve in the regulator.

(c) Case leakage. Cap the outlet port of the regulator and apply an inlet pressure of 135 ± 5 psig. Submerge the complete regulator assembly in water for 5 minutes. There shall be no evidence of continuous leakage (other than that permitted in 4.5.3(b) above).



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(d) Check valve. Connect the uninstalled check valve to a regulated pressure source in the reverse flow direction. With the valve submerged, increase the pressure to 300 ± 25 psig and hold for five minutes. There shall be no evidence of external leakage or reverse flow leakage evidenced by bubbling from the check valve inlet. Reduce the pressure to 12 ± 2 inches of water and hold for five minutes. There shall be no evidence of reverse flow or external leakage. Disconnect the regulated pressure supply from the check valve outlet and connect the supply to the inlet of the check valve. Increase the inlet pressure. Cracking pressure shall be 4 ± 2 psig.

* 4.5.4 Test data sheets. All test data shall be recorded on data sheets by the supplier as specified in the contract or order (see 6.2). Data sheets shall contain, but are not limited to, the following information:

(a) Date_____

(b) Regulator serial number_____

(c) Tests

Pressure regulation

| Inlet Pressure Psig | Orifice (Valve "A") | Outlet Pressure Psig | |
|---|---|---|---|
| (250) _____ | Small | _____ | |
| (1500) _____ | Small | _____ | |
| (2700) _____ | Small | _____ | |
| (250) _____ | Large | _____ | 40 min. |

(d) Relief valve — Cracking pressure _____psig; Reset pressure _____psig; Leakage rate _____$cm^3$/min-Std

(e) Check valve — Cracking pressure _____psig

4.6 Periodic production inspection. Periodic production samples shall be selected in accordance with 4.3.2. The samples shall be tested in accordance with the following plan:

(a) The Sample of 1 unit from the first group of 30 units will be called the first Sample and will undergo the tests of Table VI for Sample 1. The sequence of tests will be as shown in Table VI.



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(b) The Sample of 1 unit from the second and third groups of 30 units will be called the second and third Samples and will undergo the tests of Table VI for Samples 2 and 3 respectively. The sequence of tests will be as shown in Table VI.

(c) The next Sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This Sample will be called the fourth Sample and will undergo the tests of Table VI for Sample 4, etc.

* TABLE VI

Periodic Test Sequence

| Tests | Method Paragraph | Sample and Test Sequence | | |
|---|---|---|---|---|
| | | 1, 4, Etc. | 2, 5, Etc | 3, 6, Etc. |
| Vibration, Ready Condition | 4.4.4.2 | | 1* | 1* |
| Acceleration, Ready Condition | 4.4.4.3 | | | 2* |
| Shock, Ready Condition | 4.4.4.4 | 1* | | 3* |
| Temperature and Humidity | 4.4.4.5 | | 2 | 4 |
| Low Temperature Operating | 4.4.4.6 | | 3 | 5 |
| High Temperature Operating | 4.4.4.7 | 2 | | 6 |
| Operational Service Life | 4.4.4.10 | | | 7 |
| Operational Test | 4.5.3 | | | 8 |

*Visual Check After Testing

4.6.1 Test Conditions. Unless otherwise specified (see 6.2) the regulator shall be subjected to the periodic tests under the conditions specified in 4.4.2.

4.6.2 Test equipment and setup. The test equipment and setup for the periodic tests shall be in accordance with 4.4.3.

*

4.6.3 Test data. Test data shall be recorded in accordance with 4.5.4, as appropriate, and maintained as part of the inspection record.



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4.6.4 Determination of defect or failure. Failure of a regulator to successfully pass a test shall be cause for withholding acceptance of all units on hand. Corrective action shall be taken in accordance with 4.3.2.

4.7 Examination of first echelon repair parts.

4.7.1 General.

* 4.7.1.1 The information presented in this section is applicable to all components assembled into the end product and to first echelon repair parts. The information is divided into the following major divisions:

(a) Parts which are manufactured in accordance with Federal or Military specification which contain adequate quality assurance provisions.

(b) Minor hardware items.

4.7.1.2 Unless otherwise specified, sampling plans and procedures set forth in MIL-STD-105, Inspection Level II shall be used in determination of acceptability of the first echelon repair parts and all components assembled into the end product. An AQL of 1.0 shall be used for major items of inspection, and AQL of 2.5 shall be used for minor items of inspection.

4.7.1.3 The verification of the quality of repair parts is mandatory and shall be performed whether the item is a part of a repair kit or is to be assembled into the end product.

* 4.7.2 Components covered by Federal and Military Specifications. The components and parts referenced within this subsection shall be inspected in accordance with the applicable specifications. For inspection of repair parts use:

| PART | SPECIFICATION |
|---|---|
| Adapter, CRES, Class 1, Type 1, Style 3, 3/16 OD Tube | MIL-F-18866 |
| Sealing Compound, Grade AV | MIL-S-22473 |

* 4.7.3 Miscellaneous hardware. The items referenced within this subsection shall be inspected as to type and quantity.

| PART | STANDARD |
|---|---|
| Balls, Bearing | MS-19060-20 |
| Wire, Safety | MS-20995-C41 |
| Packing, Preformed | NAS617-4 |



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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2). A unit package shall consist of one regulator.

5.1.1 Level A. Preservation and packaging of equipment for Level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required, shall be accomplished with Method D-1 or D-4. Preservation and packaging shall be Method IA.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be accomplished in accordance with the supplier's own commercial practice when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C, as specified. See 6.2. Shipping container shall contain one unit package.

5.2.1 Level A. The equipment, when preserved and packaged in accordance with 5.1.1, shall be overpacked in containers complying with MIL-P-116, Method II d.

5.2.2 Level B. Not applicable.

* 5.2.3 Level C. The equipment, when preserved and packaged in accordance with 5.1.3, shall be overpacked in compliance with Uniform Freight Classification Rules and National Motor Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.4 Cushioning or dunnage. Supplier shall provide sufficient cushioning or dunnage material to adequately protect the regulator from damage due to shock and vibration loads encountered during all modes of handling and transportation. Shock and vibration tests shall be in accordance with this specification.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with requirements of MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. The regulator is intended to regulate the flow of carbon dioxide ($CO_2$) gas for use as part of a torpedo fuel delivery system.



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* 6.2 Ordering data.

* 6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Whether a preproduction sample is required (see 3.2.1)

(c) Assigned activity for preproduction tests (see 4.3.1)

(d) Assigned activity for periodic production tests (see 4.3.2)

(e) Any exceptions to this specification (see 3.2.2, 3.3, 3.4, 4.4.2, 4.6.1 and 4.5.2)

(f) Required level of preservation, packaging, and packing (see 5.1 and 5.2)

(g) Any additional marking required (see 5.3)

* 6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs herein:

| Paragraphs | Data Requirement | Applicable DIDs |
|---|---|---|
| 4.4.5, 4.5.4, 4.6.3 | Test data sheets | - |

DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Definitions

6.3.1 Ready. The condition of the regulator when fully assembled after completion of servicing and maintenance, with the ports plugged.

6.3.2 Operating. The condition of the regulator when operating.

6.3.3 Nonoperating. The condition of the regulator when stored, transported, or handled. Tests on the regulator, in the nonoperating condition, shall be performed without packaging or packing.



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6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in Hertz.

6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in gravitation units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration.

Shock, in general, can be described as a buildup to 90 percent of peak force in less than 0.010 second. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds and shock wave form equivalent to one half sine wave. Duration is defined as the interval between the point of zero amplitude, from which the shock wave form builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the regulator is exposed.

6.3.8 Humidity. Humidity is the relative humidity of the environment to which the regulator is exposed, expressed in percent.

* 6.5 Storage life. The regulator, packed and packaged in accordance with section 5, herein, has a storage (shelf) life of 5 years and, thereafter, will meet the performance requirements of this specification.

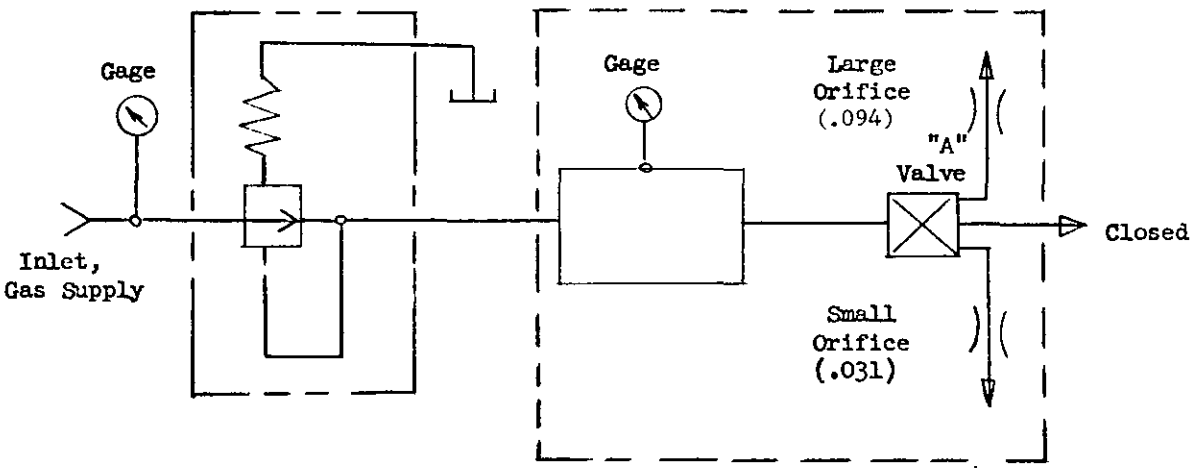
* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal and notations and relationship to the last previous issue.

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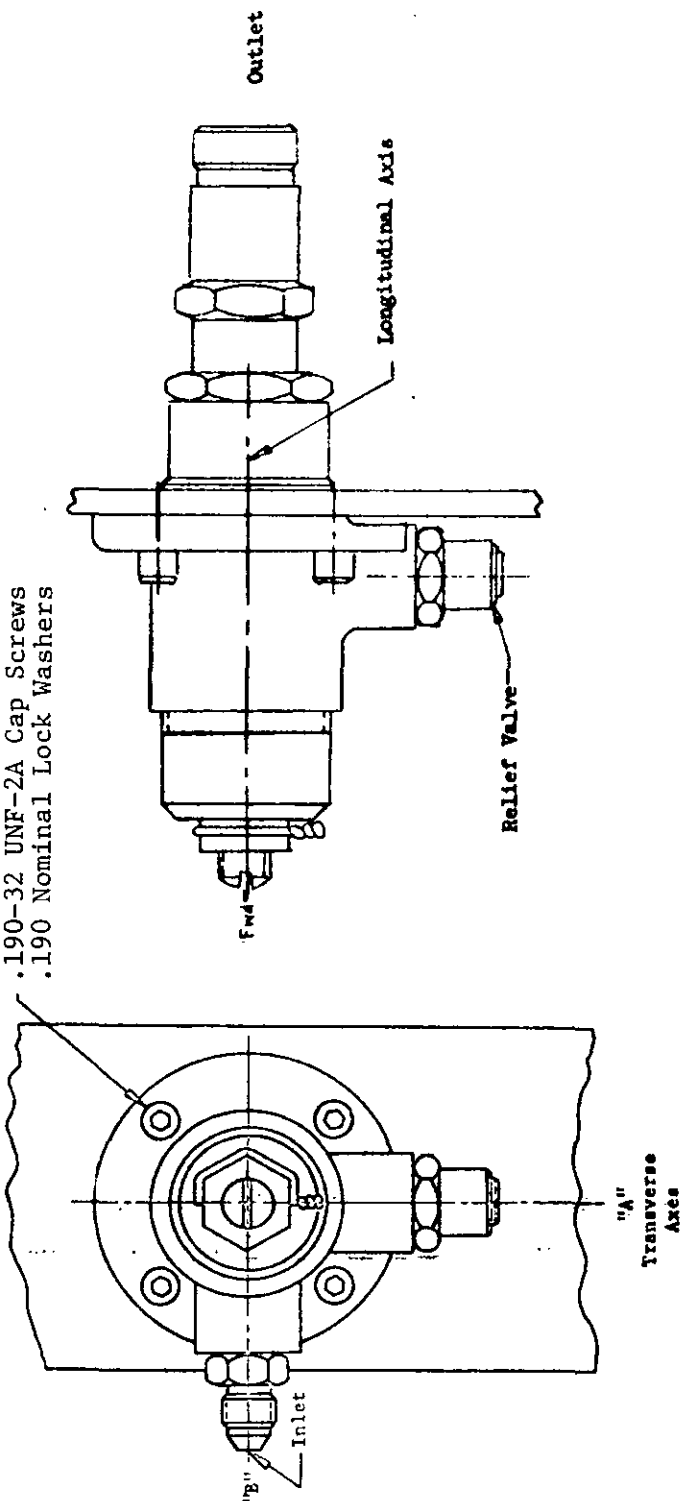
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* Figure 1. Regulator Test Setup



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* Figure 2. Regulator Mounting



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-82441A(OS), Regulator Assembly, Carbon Dioxide Pressure

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

**DD FORM 1 JAN 66 1426** REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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